



NATIONAL OPEN UNIVERSITY OF NIGERIA

PLOT 91, CADASTRAL ZONE, NNAMDI AZIKIWE EXPRESSWAY, JABI-ABUJA

**FACULTY OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE**

2024_2 EXAMINATION

COURSE CODE: CIT308

COURSE TITLE: FORMAL METHODS AND SOFTWARE DEVELOPMENT

COURSE CREDIT: 3 UNITS

TIME ALLOWED: 2 HOURS

INSTRUCTION: ANSWER QUESTION 1 AND ANY OTHER THREE QUESTIONS

Q1

- (a) Mention 8 characteristics for system description **4 marks**
- (b) Draw a level 0 data flow diagram for hospital management system? **5 marks**
- (c) Construct a truth table for the formula $\neg(x \vee y) \leftrightarrow ((\neg x \wedge y) \vee z)$ **6 marks**
- (d) Discuss the 4 types of software **maintenance** you know? **8 marks**
- (e) List 4 software requirement characteristics? **2 marks**

Q2

- (a) Suppose x is a real number. Consider the statement "If $x^2=4$, then $x=2$." Construct the converse, the inverse, and the contrapositive. Determine the truth or falsity of the four statements (1) the original statement, (2) the converse, (3) the inverse, and (4) the contrapositive; using your knowledge of algebra. **8 marks**
- (b) Discuss the NOT and IF AND ONLY IF connectives? **6 marks**
- (c) What advantages do structure charts have over DFD? **1 mark**

Q3.

- (a) Discuss the unit, system and user acceptance testing? **6 marks**
- (b) Mention 3 types of formal methods in Software Development? **3 marks**
- (c) List 6 application specific GUI components **6 marks**

Q4

- (a) Tabulate the strength and weakness of the water-fall model? **6 marks**
- (b) Using a diagram, highlight the data flow diagram symbols **3 marks**
- (c) Mention 6 characteristics for system description **3 marks**
- (d) Distinguish between modular programming and structured coding? **3 marks**

Q5

- (a) Given: $A=\{1, -2, 5, 7, 9, 10\}$; $B=\{2, 3, 5, 8, 3\}$ and $C=\{5, 4, 6, 8, 10\}$. Describe and provide results for $(A \cup B) \cap C$ and $(B \cup C) \cup A$ **8 marks**
- (b) Discuss formal specification as a precursor to software development? **3 marks**
- (c) Discuss the importance of software documentation? **4 marks**

Q6.

- (a) Why do we study software development? **2 marks**
- (b) Outline any 6 benefits of using formal methods in software development? **3 marks**
- (c) With a diagram, describe the distinction between HIPO and IPO chart? **10 marks**